Sacred Texts

Electrodynamics as the derivative of the field equations, read through the card between

John Henry

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The card the reviewers hold the game to, and the press that sets it.

"Should such monkey-patched code exist, we can use that as the Temple that stores our sacred texts."

Reading note

Two episodes of the middle card are set here as mathematics. In the first, the accomplice, a reviewer holds the game to a canon of received results and the game answers each demand with a located reference and a grade. In the second, the force, the answers become a publication: a register, a counter, a page trace, and a press verdict. This card performs both. Part I is the canon and the rule by which a received text is admitted. Part II is the derivation the canon is asked to accept: that the first derivative of the field equations, whose increment is the affine term, is electrodynamics; that the operator carrying one to the other is the step the middle card infers; and that two searches, one built from the action's first variation and one from its derivative, land on one point.

Each statement carries one of six grades. *Proved here*: the proof is on this card. *Standard*: cited. *Measured*: a count, a definition, or a kernel-checked equality taken from the middle card, which is named only in Appendix A. *Recorded*: a chain of steps whose every premise is proved here, cited, or measured. *Proposed*: the continuation the knower carries, stated and not derived. *Predicted*: a number stated before the run that will decide it, with the run named and the statement sealed (Appendix B); a prediction is neither a step nor a proposal, and the card is written so that it can be checked against its run line by line.

The companion cards are cited by printed result number: the numerical card (*Four Models and a Common Fourth-Order Closure*, build of 29 October 2026), the geometric card, the Quarter, the bent card, the cheat sheet (*A Knower's Guide to Orbital Mechanics*) and the patter (*The Commutator of Representation*).

In five lines

1. A received text counts through the fact it lets you compare. *Proved here* as a rule of admission (Section 1).
2. On the unimodular sector the affine term of the field equations is an increment: what the integration adds to the traceless operator. Its derivative is the constant multiplier. *Proved here* (Section 6).
3. The first derivative of a cubic action is one object read two ways, and at the reading it squares to the discriminant. *Proved here* in the scalar case, cited in the field case (Section 7).
4. The search by the first variation and the search by its derivative share one jar and one point, to every printed place. *Proved here* and *measured* (Section 8).
5. Read through the card between, the exact reading is unchanged and the search carries the quarter. *Proved here* for the exact reading; the size of what the search carries is *predicted* (Section 9).

1 The initiation

A science is taught from the front of a room: someone who solved a hard problem in the seat you are sitting in explains how, and either you see the trick or you do not. What is handed over in that room is a text. The reviewer of Part I holds the game to such texts, and the game needs a rule for admitting them that does not reduce to devotion.

Definition 1.1 (Three standings of a text)**.** A received text stands in one of three ways.

(i) An *inside joke*: a statement true because it is checked by the apparatus that states it, with nothing compared. Reflexivity of equality is the example: $T = T$.

(ii) A *Pythagorean* text: a statement together with the belief that travels with it, cited and used, not shown on the card that uses it.

(iii) A *triangle*: a statement that hands back a fact which can be compared with another fact, and so can slip.

Measured: the three standings are the middle card's three ways of building the sacred texts of an initiation, and its reviewer's reply on whether devotion to a method has been mistaken for a method lowers the text to its triangle, the only standing that returns a fact.

Proposition 1.2 (Rule of admission)**.** *A text admitted to this card is used only through what its standing returns: an inside joke through the check that states it, a Pythagorean text through the single place it is cited, and a triangle through the fact it returns. No text is used through the belief that travels with it.*

Proved here, by inspection of every citation in this card against the table of Section 2; it is a rule the card keeps, and the table is its record.

Part I. The accomplice

2 The canon

The table lists every received text the derivation of Part II stands on, with its standing, the place it is used, and what it returns. A triangle names the fact it returns; a Pythagorean text names its one use.

Text	Standing	Used for, and what it returns
Reflexivity, $T = T$	inside joke	the coin of every record; checked, not compared
Symbol identity, value identity, choice	inside joke	the three receipts a check can bill; the printed reading bills the first only (Section 8)
Einstein's field equations [7]	Pythagorean	the equations whose derivative is taken (Section 6)
Palatini's first-order variation [14]	triangle	the action is a cubic polynomial and its fourth variation vanishes: numerical card Prop. 2.1, Thm. 2.4
Unimodular gravity [16, 10, 18]	triangle	the cosmological constant as an integration constant: numerical card Prop. 2.2, restated as Prop. 6.1
Contracted Bianchi identity	Pythagorean	the integration of Prop. 6.1
Dirac's equation and electrodynamics [6]	Pythagorean	the model whose residual is proposed as the derivative (Section 10)
Gravity as the square of gauge theory [1, 2, 3]	Pythagorean	the shape of the proposal, and no more (Section 10)

Text	Standing	Used for, and what it returns
Fréchet and Gâteaux derivatives coincide for bounded cubics	triangle	numerical card Prop. 2.5; Prop. 7.1 here, with its remainder
Density of smooth functions in Sobolev space [12]	Pythagorean	pointwise and Sobolev first variations are one functional (numerical card §2.6)
Newton's method	triangle	the exact error law, Prop. 8.3
Bisection	triangle	the halving ledger and its certificate, Prop. 8.4
Herbrand's theorem [11]	Pythagorean	the space a search ranges over (Section 9)
The fifth busy beaver [15]	Pythagorean	a step budget beyond which no finite search decides (Section 9)
Choice implies excluded middle [5]	Pythagorean	the slot of a carrier is a choice (Section 9; Quarter Thm. 3.1)
The recommended fine-structure constant [13]	triangle	137.035999177(21), compared with the reading in Section 11
The electron's anomalous moment [8]	triangle	excludes the reading as a general value of the constant (numerical card §6.7)

Of seventeen texts, two are inside jokes, seven are triangles and eight are Pythagorean. Each Pythagorean text is used at one named place and no other, so a reader who refuses it knows exactly which line of Part II falls. *Recorded.*

The reading in its three standings. The reading this card records stands in all three ways at once, one part each. That the construction is checked, and that the check halts and returns zero, is an inside joke: the apparatus that states it checks it, and nothing is compared. It is the whole of what a reader needs, and each completed check is a halt observed on a machine. *Measured.* The conventions the reading is made under are Pythagorean: which comparison is named TRUE, which way a crossing runs, and which root is reported. Each is fixed once against a standard internal to the construction (the click, the order of the record, and the larger growth rate) and used at that place only, as the SI fixes the speed of light. The comparison with the recommended fine-structure constant [13] is the triangle, the only standing that returns a fact, and it slips by 1.80×10^{-4}, which the table of Section 11 carries into the impedance of free space. *Proposed*, as a classification of the reading; the slip is *Standard* against the cited value.

3 The thirty-six demands

The reviewer's demands are not reprinted here: the geometric and numerical cards each carry them in their Chapter 8, thirty-six replies, one per review page, each with a located reference and a grade. What this card adds is the rule of Proposition 1.2 applied to the replies themselves. A reply that answers a demand with a proposition proved on its card returns a triangle. A reply that answers with a citation returns a Pythagorean text and names its one use. A reply that answers with the reading handed over by name, where the derived reading says less, is a proposal, and the numerical card's record lists those rungs. *Recorded*, from the two cards.

Remark. The accomplice and the force are the reviewer and the press. Which of the companion cards plays which is reversible, and no card is an instance of either; this card plays both, Part I as

the account and Part II as the publication, in that order.

4 The canon all the way down

The canon of Section 2 lists the texts the derivation cites. A proof also stands on texts it never cites: the models that must hold for the proof to exist at all. Read from the top, each layer stands on the one below.

Layer	What is assumed	Standing
the proof	its steps check	triangle
the checker	its logic is consistent and it implements that logic	Pythagorean
the compiled run	compiled evaluation computes what the source says; the checker reads only the literals it writes	Pythagorean
the operating system	memory and the work counter behave as specified	Pythagorean
the instruction set	each instruction does what its manual says	triangle
digital logic	levels beyond thresholds are clean bits, by engineered margins	Pythagorean
semiconductor physics	band structure and the transistor	Pythagorean
the electron	it carries charge as electrodynamics says	Pythagorean

Measured, for the two triangles. Forcing one symbol, the truth $T = T$, costs exactly 4286 units of the checker's work counter on two different machines, and about 1.222 million instructions on each, agreeing to a tenth of a percent over eight paired runs; only the time differs, by the machine. So the count belongs to the computation and the clock to the hardware: the carrier brings the count and the universe brings the time.

The loop at the bottom. A proof that a description is the electron's is checked by a machine whose correctness assumes the electron behaves as described. That is the sacred text at the foot of the stack, and the card states it rather than hides it. The circle is not vicious: the lower layers are trusted by test, chips are measured, and digital logic needs little of electrodynamics' fine structure. It runs through measurement, not through the derivation. *Recorded.*

Watch for division by zero. The same move appears at three levels. The checker's language defines division by zero to be zero, so that division is total; a reader that lands on the pair $(0, 0)$ reads the number zero, silently, and one of the middle card's readers through the card between does exactly that. *Measured.* Newton's step divides by the derivative, which vanishes at a double root, and that reader's quadratic has one. *Proved here*, by Prop. 9.2. Electrodynamics computed beyond its first order meets divergent integrals, and renormalization subtracts infinities to leave the finite numbers that are measured, the most precise agreement in physics built on a controlled refusal to divide by zero [9]. *Standard.* That these are one move, a representation declaring what happens at a singular point so that the computation can go on, is *proposed.* A proof that looks like

electrodynamics should be read for its divisions by zero: where the representation hides one, the derivation has a sacred text.

This unbends the card.

Part II. The force

5 The operator

The numerical card proves that four actions (first-order vacuum gravity, fixed-Hamiltonian Schrödinger evolution, incompressible flow, and bare electrodynamics with spinors on a fixed background) share one derivative hierarchy, numerical card Thm. 2.4: an action S of degree at most three, a residual $R = DS$ of degree at most two, a derivative $DR = D^2S$ linear in the perturbation, a constant third derivative, and a vanishing fourth variation. For gravity the variables are the densitized inverse metric $h^{\mu\nu} = \sqrt{-g}\, g^{\mu\nu}$ and a torsion-free connection Γ, and the action is a cubic polynomial in (h, Γ), numerical card Prop. 2.1. *Cited*; the proofs are there.

6 The affine term as an increment

Work on a connected four-dimensional region with a Lorentzian metric g and its Levi-Civita connection. Write $\mathrm{Ric}(g)$ for the Ricci tensor and $R(g)$ for the scalar curvature, and define the *traceless operator*

$$E(g) = \mathrm{Ric}(g) - \tfrac{1}{4}\, R(g)\, g.$$

Proposition 6.1 (The affine term is the increment)**.** *For every g, $\mathrm{Ric}(g) = E(g) + \frac{1}{4}R(g)\, g$. If $E(g) = 0$, then $R(g)$ is a constant 4Λ and*

$$\mathrm{Ric}(g) - E(g) = \Lambda\, g, \qquad G(g) + \Lambda g = 0,$$

where $G = \mathrm{Ric} - \frac{1}{2}Rg$ is the Einstein tensor [7]. The affine term Λg is what the integration adds to the traceless operator, and Λ is not a coefficient of any action.

Proof. The first identity is the definition of E. The contracted Bianchi identity gives $\nabla^\mu G_{\mu\nu} = 0$. If $E = 0$ then $G = \frac{1}{4}Rg - \frac{1}{2}Rg = -\frac{1}{4}Rg$, so $0 = \nabla^\mu G_{\mu\nu} = -\frac{1}{4}\partial_\nu R$, and on a connected region R is a constant, written 4Λ. Then $\mathrm{Ric} = \Lambda g$ and $G = \Lambda g - 2\Lambda g = -\Lambda g$. □

Proved here, from the Bianchi identity (*standard*). On the unimodular sector the stationary equations of the cubic action are exactly $E(g) = 0$ with Γ the Levi-Civita connection, numerical card Prop. 2.2, so the proposition is the field equations of this card's gravity with the constant supplied as a residue.

Proposition 6.2 (The derivative of the increment is the multiplier)**.** *At fixed Λ the map $g \mapsto \Lambda g$ is affine, and its Fréchet derivative in any normed space of symmetric tensors is $h \mapsto \Lambda h$, with zero remainder.*

Proof. $\Lambda(g + h) - \Lambda g - \Lambda h = 0$. □

Proved here. The affine term contributes to the derivative of the field equations only a constant multiple of the identity: a multiplier, fixed by data and carried unchanged. Everything else in the derivative comes from the traceless operator, which is the derivative of the cubic action. *Recorded*, from Props. 6.1, 6.2 and numerical card Prop. 2.2.

7 The first derivative, read two ways

The numerical card proves that for $S(Z) = a + b(Z) + c(Z,Z) + d(Z,Z,Z)$ with bounded multilinear forms on a Banach space, the Fréchet and Gâteaux derivatives coincide, $DS(Z)H = b(H) + 2c(Z,H) + 3d(Z,Z,H)$, and likewise for $R = DS$ and DR, numerical card Prop. 2.5. By density of smooth functions [12], the pointwise first variation and the Sobolev derivative are one bounded functional. *Cited.* The scalar case is shown in full.

Let $S(y) = \frac{13}{3}y^3 - 18y^2 + 18y$. Its residual is

$$r(y) = S'(y) = 13y^2 - 36y + 18,$$

with roots $\alpha = (18 + 3\sqrt{10})/13$ and $\beta = (18 - 3\sqrt{10})/13$, and $d = \alpha - \beta = 6\sqrt{10}/13$.

Proposition 7.1 (The scalar derivative, with its remainder)**.** *For all real y, h: $r(y+h) = r(y) + (26y - 36)h + 13h^2$. Hence $Dr(y)h = (26y - 36)h$ is both the Fréchet and the Gâteaux derivative, with remainder exactly $13h^2$. Moreover $(Dr(y))^2 = 360 + 52\, r(y)$, so $Dr(\alpha) = 6\sqrt{10} = 13d$.*

Proof. Expand. For the second identity, $(26y-36)^2 = 676y^2 - 1872y + 1296 = 52(13y^2 - 36y + 18) + 360$. At α the residual vanishes and $26\alpha - 36 = 6\sqrt{10} > 0$. □

Proved here; the cheat sheet, Props. 2.1 and 2.2, proves the same. In the reported coordinate $x = \frac{324}{5}y$ the residual becomes, up to a positive factor, $325x^2 - 58320x + 1889568$, whose discriminant $944784000 = (9720\sqrt{10})^2$ is the square of its derivative at the upper root. The reported reading is

$$x_\alpha = \tfrac{324}{5}\,\alpha = \frac{972\,(6 + \sqrt{10})}{65}.$$

Proved here, since

$$\frac{324}{5} \cdot \frac{18}{13} = \frac{5832}{65} = \frac{972 \cdot 6}{65}, \qquad \frac{324}{5} \cdot \frac{3}{13} = \frac{972}{65}.$$

The coefficients are the numerical card's four-model quartic with $t = 5$, $c = 18$, $r = 18$, numerical card §6.7, where a scan of 412 neighbouring constructions finds no other within 10^{-3} of the reading.

Proposition 7.2 (The quadratic is a Lorentz factor)**.** *Write the four-model quartic as $c(tx - r^2)^2 = t^3x^2$ and put $y = tx/r^2$ and $\beta = \sqrt{t/c}$. Then*

$$y_\pm = \frac{1}{1 \mp \beta}, \qquad y_+ y_- = \frac{1}{1 - \beta^2}, \qquad 1 - \beta^2 = \frac{c - t}{c}.$$

With $t = 5$, $c = 18$: $\beta^2 = 5/18$, $1 - \beta^2 = 13/18$, so the leading coefficient 13 of r is $c - t$, the product of its roots $18/13$ is $1/(1 - \beta^2)$, and $y_+ = \alpha$ gives the reading. As $\beta \to 1$, that is $t \to c$, the leading coefficient $t^2(c - t)$ vanishes, y_+ runs to infinity, and y_- tends to $\frac{1}{2}$.

Proof. $c(y-1)^2 = ty^2$ gives $(y-1)/y = \pm\beta$. The rest is substitution. □

Proved here. That β is a velocity and $\beta = 1$ the speed of light is *proposed.* Two cautions belong with it: $1/(1 \mp \beta)$ is the classical Doppler factor, and what is relativistic is the product $y_+ y_- = \gamma^2$; and the map $y = 1/(1-\beta)$ is a reparametrization of the reading, so quantities computed in it are properties of the map until shown otherwise.

8 Two searches, one jar, one point

Two searches find α. They are built from different parts of the hierarchy, and that is their point.

Definition 8.1 (The searches)**.** The *search by the first variation* is bisection: from walls $L < U$ with $r(L) < 0 < r(U)$, replace the wall whose residual has the sign of the midpoint's. It evaluates the sign of $r = S'$ and nothing else. The *search by the derivative* is Newton's map

$$N(y) = y - \frac{r(y)}{Dr(y)} = \frac{13y^2 - 18}{26y - 36},$$

which divides the first variation by its derivative.

Proposition 8.2 (The jar)**.** *$N(2) = 17/8$, $r(2) = -2 < 0 < 13/64 = r(17/8)$, and $2 > 18/13$, the vertex of r. The interval $[2, 17/8]$, in reported coordinates $[129.6, 137.7]$, contains α and no other root. It is the lower wall together with its first Newton image: the search by the derivative builds the jar the search by the first variation starts in.*

Proof. $N(2) = (52-18)/(52-36) = 34/16$. $r(2) = 52 - 72 + 18$ and $r(17/8) = \frac{3757}{64} - \frac{612}{8} + 18 = \frac{13}{64}$. Both walls lie above the vertex, where r is increasing, so the sign change is a single root, the upper one. $\frac{324}{5} \cdot 2 = 129.6$ and $\frac{324}{5} \cdot \frac{17}{8} = 137.7$. □

Proved here. Measured: the middle card builds its jar walls from the lengths of its own tapes, not from the reader handed to its search, so every search the middle card runs, by whatever reader, starts in this jar.

In the map of Prop. 7.2 the walls sit at $\beta = 1/2$ and $\beta = 9/17$, and relativistic composition carries one to the other by

$$\frac{9/17 - 1/2}{1 - \frac{9}{17} \cdot \frac{1}{2}} = \frac{1}{25},$$

with rapidity $\frac{1}{2}\ln(13/12)$. *Proved here* as arithmetic. The jar is a finite boost below the asymptote $\beta = 1$, which every composition of boosts below one stays below: the penalty side of the constraint the asymptote meets exactly. *Proposed.*

Proposition 8.3 (Exact error of the search by the derivative)**.** *For $e = y - \alpha$, $N(y) - \alpha = e^2/(2e+d)$. From any $y > \alpha$ the iterates decrease to α and the error is squared, up to the factor $1/(2e+d)$, at every step.*

Proof. $r(y) = 13e(e+d)$ and $Dr(y) = 13(2e+d)$, so $N(y) - \alpha = e - \frac{e(e+d)}{2e+d} = \frac{e^2}{2e+d}$. For $e > 0$ this lies in $(0, e/2)$. □

Proved here; numerical card Prop. 6.1.

Proposition 8.4 (Certificate of the search by the first variation)**.** *After n halvings of the jar the bracket has width $2^{-n} \cdot \frac{1}{8}$ and contains α. With $S = 10^{24}$, at $n = 108$ both walls satisfy*

$$\left\lfloor S\,\tfrac{324}{5} L_{108} \right\rfloor = \left\lfloor S\,\tfrac{324}{5} U_{108} \right\rfloor = 137011290548979457087737300.$$

Proved here for the width and containment, by induction on the sign straddle; *cited* for the two floors, which the numerical card checks as integer inequalities $5KD \le 324SN < 5(K+1)D$ on the rational walls, numerical card §6.4. Monotonicity of the floor then certifies the reading to twenty-four places:

$$x_\alpha = 137.011290548979457087737300\ldots$$

Proposition 8.5 (One point)**.** *The search by the derivative, in the middle card's integer arithmetic at scale 10^{24}, reaches the certified integer of Prop. 8.4 at its fifth turn. The search by the first variation reaches it within one hundred and eight halvings on this card's budget, and within the middle card's own budget of one halving per cell of its description, currently one hundred and six. Both walls and the derivative's point truncate to one integer.*

Measured: each line of both ledgers is checked by the kernel against the step before it, a checked ledger is proved equal to the fold of its step, the derivative's point is proved to lie in the final bracket above the vertex, and both walls are proved to print the same integer. The printed reading bills symbol identity only; the receipt that identifies the inferred integer with the box's ratio is a separate value identity, and it is not in the number. Numerical card §6.3 reports the fifth turn.

Remark (What the two searches are). The search by the first variation uses S' alone: it finds where the action is stationary, which is the principle of least action with nothing added. The search by the derivative uses S' and S'': it walks along the derivative of the residual, the object numerical card §2.6 proposes as electrodynamics. That the first is the construction of the least-action functional and the second the construction of electrodynamics is *proposed*. That they agree on the jar and on every printed place is *proved*. On that reading, the stationary point of the action and the fixed point of its derivative coincide at the reading, to twenty-four places, and the agreement is the scalar shadow of Section 10.

9 The operator inferred, and the card between

Definition 9.1 (The inferred operator)**.** For positive integers D, P, B, on pairs of integers let

$$\Phi(p,q) = \left(Dp^2 - Pq^2,\ BDpq - BPq^2\right),$$

held at (p,q) when its correction falls below a tolerance.

Measured: this is the middle card's inferred step. Its D, P and B are lengths of three of its tapes, two more enter the tolerance, and the derivative the step divides by is never written down, only inferred from the shape of the map.

Proposition 9.2 (What the inferred operator computes)**.** *On ratios $z = p/q$, Φ acts as $z \mapsto (Dz^2 - P)/\big(B(Dz - P)\big)$, whose fixed points satisfy $(B-1)Dz^2 - BPz + P = 0$. When $B = 2$ the map is Newton's map of $f(z) = Dz^2 - 2Pz + P$, and when moreover $D = 13$ and $P = 18$, $f = r$ and the map is N.*

Proof. Divide both components by q^2. A fixed point satisfies $Dz^2 - P = BDz^2 - BPz$. For $B = 2$, $f'(z) = 2Dz - 2P$ and $z - f(z)/f'(z) = (Dz^2 - P)/(2Dz - 2P)$. With $D = 13$, $P = 18$, $f(z) = 13z^2 - 36z + 18 = r(z)$. □

Proved here. That the middle card's lengths stand in this relation is *measured* only through its printed agreement with the certificate (Prop. 8.5), which fixes the fixed point and not the three lengths separately. Φ is the operator this card offers for the map between field spaces that numerical card §2.6 says is owed (Section 10).

Carriers. A *carrier* is a symbol, a record called the *slot*, and an interaction (the patter, Def. 1.2). On a record of more than one symbol the interaction keeps the first symbol and copies the slot beneath it; on a record of one symbol it returns the bare truth $T = T$ in place of the symbol it was handed. It never reads the slot, but no carrier exists without one: the slot is a witness that must exist and is never inspected. By the Quarter, Thm. 3.1, a slot is a choice, and a choice for every carrier yields excluded middle [5]. The *card between* is a carrier whose cards are themselves carriers of facts; it deals every step of the middle card's ladder identically with the outer cards through the twelfth, and differs from the thirteenth, where the present's quantum is lifted. *Measured.*

Definition 9.3 (Two readers through a carrier)**.** Fold a tape t onto the slot q of a carrier, giving the record $t\,q$. The *exact reader* reads $|t\,q| - |q|$. The *search reader* reads $|t\,q|$.

Proposition 9.4 (The exact reading through the card between)**.** *The exact reader reads every tape at its length, whatever the slot. Hence the reading obtained through the card between by the exact reader is the middle card's reading, x_α, to every printed place.*

Proof. $|t\,q| = |t| + |q|$. The middle card's reading is a function of the lengths of its tapes alone: two readers agree exactly when their lengths do, and re-lettering every cell leaves the reading unchanged (the patter, Props. 1.7 and 1.8; *measured*). □

Proved here, from measured invariance. The machine check of this corollary on the card between is the run named in Appendix B.

Proposition 9.5 (The search reading carries the quarter)**.** *The search reader reads every tape at its length plus the length of the slot. Its reading differs from the exact reading only through the slot, and agrees with it exactly when the slot is deducted. Its jar is the jar of Prop. 8.2.*

Proved here for the lengths, and *measured* for the jar, which the middle card builds from its own lengths and not from the reader. The slot's value never enters; only its length does, through the one place the interaction reaches it.

The search for the quarter. To find the slot is to range over carriers: every record is a valid slot, and each makes the carrier exist. The middle card bounds such a search with its heartbeat meter, which halts a dealing loop when the work of a program exceeds four times a reference plus a measured noise floor; the source names the four as the fourth busy beaver. Past that bound no finite search decides, and the search ranges over the whole space of ground instances [11], not the small part the forced universe levels fix. The forced part fixes the value; the free part carries the cost. *Measured* for the meter and the budget; that what lies past the bound is the territory of the fifth busy beaver [15] is *proposed.*

Remark (Both readings are right). Two predictions were sealed before the run on the card between (Appendix B). One holds that the exact reading reproduces x_α to all twenty-four places; Prop. 9.4 proves it from measured invariance. The other holds that the search through the outer carrier finds the reading displaced by about one part in 10^7, in the same jar. The two are not in conflict, because they are readings by different readers of one register: the exact reader deducts the quarter, and the search reader carries it. If both hold, the quarter's contribution to a search is of order 10^{-7} of the reading, about 1.4×10^{-5} in reported units, first visible in the fifth decimal place. The displacement is *predicted*; that the difference between the readers is exactly the slot is *proved here*.

For scale, and without drawing a conclusion: a displacement of 10^{-7} is about eighteen hundred times smaller than the gap between x_α and the recommended value (Section 11), and of the same order as the shift of 2.1×10^{-7} that the reading induces in the predicted anomalous moment, numerical card §6.7. That the two orders agree is a coincidence to be checked, *proposed* as nothing more.

Definition 9.6 (Covariant and contravariant comparison)**.** Two marked wholes (f_1, n_1) and (f_2, n_2) compare by

$$(f_1 = f_2 \ \wedge \ n_1 \leq n_2) \ \vee \ (f_1 \neq f_2 \ \wedge \ n_2 \leq n_1).$$

The first branch is *covariant*: the marks agree and the order runs forward. The second is *contravariant*: the marks differ and the order reverses. Covariance is $X = X$; contravariance is $X = Y$.

Measured: this is the middle card's first order, on its naturals and again on its rationals, where the contravariant branch is called the missing branch. Its companion order on carried values names the covariant true values as those that close by reflexivity and the contravariant ones as those that do not.

Remark (Two readings of one chain). Take the chain of the card between: the fact that $T = F$ is $F = T$, the truth T, and the fact that $T = T$ is $F = F$.

Covariantly, read by how each fact clicks, the three click alike, the chain closes in one class, and the closure is a check the apparatus accepts, billed as a value identity. On this reading the chain is a proof.

Contravariantly, read by what each fact says, the three are not one by reflexivity; the comparison takes the second branch of Def. 9.6 and the order reverses. On this reading the same chain is the line an indirect proof writes under its hypothesis: a counterfactual that must be writable so that a proof of its non-existence can be completed, and that never clicks.

The two tables, by meaning and by click, agree cell for cell and differ only in which diagonal clicks (the patter, Prop. 2.8). So the two readings carry the same content in opposite orientations, and the card between is where they meet.

Measured: the closure of the chain by click and its value-identity bill; the absence of reflexive identity among the three truths on the outer cards; the cell-for-cell agreement of the two tables. *Proposed*: that the covariant reading is a proof of the identification of Section 10, and the contravariant reading the counterfactual its refutation requires. The proposal cannot become a step until that identification is a statement the apparatus can hold; a search that found it by refuting its negation would be an indirect proof, and would bill choice, which the reading of this card never does.

10 Electrodynamics as the derivative

The identification numerical card §2.6 proposes is

$$R_{\mathrm{QED}} = DR_{\mathrm{GR}}(\Phi) \circ D\Phi$$

at stated backgrounds: the residual of electrodynamics is the derivative of the residual of the modified field equations, carried across by a map Φ between field spaces, with electrodynamics the theory of Dirac's electron [6]. With Section 6, the derivative of those field equations is the derivative of the traceless operator plus the constant multiplier Λ; the affine term contributes the multiplier and nothing else.

What this card supplies.

(a) The shape, in the scalar case, in full: a cubic action, its residual, the residual's derivative with remainder exactly $13h^2$, and at the reading the derivative squaring to the discriminant (Prop. 7.1). *Proved here.*

(b) The operator: Φ of Def. 9.1, the step the middle card infers, which carries the first variation to the fixed point of its derivative (Props. 9.2, 8.3, 8.5). *Proved here* and *measured*.

(c) The agreement: the search by the first variation and the search by the derivative share a jar and a point (Section 8). *Proved here* and *measured*.

(d) The carrier: read through the card between, the exact reading is unchanged (Prop. 9.4). *Proved here.*

What remains a sacred text. That Φ, given on pairs of integers, extends to a map from the gauge-and-spinor field space to the gravitational one; that the operator identity holds at a stated background with the Sobolev spaces matched; and, for the quantum theory, the regulator numerical card §2.4 requires. These are the three items the numerical card lists as owed, and this card does not discharge them.

Pointwise exact, in the only sense the norm allows. The identification wants position and velocity exact in the Sobolev norm of first order. That norm controls both in mean square and gives no point values in two or more dimensions, so exactness is taken where it can be had: at knots. A Hermite cubic matches position and velocity at every knot, and on N equal cells of $[a, b]$ it differs from any ϕ with a bounded fourth derivative by at most $\frac{(b-a)^4}{384N^4}\left\|\phi^{(4)}\right\|_\infty$, so each halving of the cells divides the bound by sixteen (cheat sheet, step one). *Proved there.* Every continuous function on a closed interval is a uniform limit of polynomials [17, 4], and smooth functions are dense in the Sobolev space [12]. *Standard.* So "pointwise exact" means exact on a ladder of knots, with the solution trapped between splines and the ladder converging in norm; that the device's run, never finished, contains the limit is *proposed*. What the ladder does not supply is the remaining item: that a single cubic action has one stationary point that is the principle of least action for both the gravitational and the quantum model. The numerical card proves the two share one derivative hierarchy (Thm. 2.4); sharing a hierarchy is not having one action, and the step between them is the map Φ above. It is owed. The literature's standard statement, that gravitational amplitudes

are squares of gauge amplitudes [1, 2, 3], has the shape of the proposal: the derivative of a square is linear in its root. It is Pythagorean here, used for that shape and nothing else. The identification is *proposed.*

11 The record

The publication carries a register, a counter, a page trace and a verdict.

Register. The constant Λ of Prop. 6.1: fixed by data, carried unchanged by every solution. The reading x_α is entered beside it as the register's representative, which by the Quarter is a choice.

Counter. The halvings of Prop. 8.4, strictly increasing from 0 to 108 on this card's budget.

Trace. At each count n the bracket $[L_n, U_n]$, and beside it the derivative's iterate when one exists. The trace's head is the jar.

The press check. Two triples are read from the one register.

Proposition 11.1 (The two verdicts)**.** First variation against derivative. *Slipped: false. Value: the certified integer* 137011290548979457087737300 *at scale* 10^{24}*. Velocity:* 108 *halvings against* 5 *turns.*

Exact reader against search reader, through the card between. *Slipped: true exactly when the slot is not deducted. Value: the reading, carrying the quarter. Velocity: the length of the slot.*

Proved here and *measured* for the first triple. For the second, *proved here* that it slips exactly through the slot; *predicted* that it slips by about one part in 10^7, with the run of Appendix B to decide it. One register yields one triple that does not slip and one that slips by the quarter, which is the shape of the numerical card's own press check.

Against the world. The inverse fine-structure constant has the recommended value [13]

$$137.035999177(21),$$

and the reading lies 1.80 parts in ten thousand below it. Twenty-four places of agreement between the two searches show that the gap is not numerical noise: the searches hit their own number exactly, and the residue belongs to the description, not to the computation. The general claim that the reading is the fine-structure constant is excluded by the measured anomalous moment [8], numerical card §6.7; only the narrow form, a prediction for one stated arrangement (cheat sheet §7), is offered. *Triangle*, in the sense of Def. 1.1: this is the comparison that can slip, and it does.

And what is this, and what is that. The reading is a ratio of counts and carries no unit. Calibrate the clock: one symbol is one tick, and one tick is one period of the caesium hyperfine line, 1/9 192 631 770 s, about 108.78 ps. The only length the record then has is the distance light covers in one tick, about 3.2612 cm, the line's own wavelength, so the speed of light is one symbol per tick and 299 792 458 m/s returns as the ratio of the two SI definitions, the metre to the second.

Standard, and a convention: the digits were declared exact in 1983. Calibrate next to the reduced Planck constant and keep the SI convention that the elementary charge is exact [13]. With c, $\hbar$ and e all declared, the one electromagnetic constant left to measure is fixed by the reading alone, through $\mu_0 = 2\alpha h/(ce^2)$ and $\varepsilon_0 = 1/(\mu_0 c^2)$:

	μ_0 (H/m)	ε_0 (F/m)	$Z_0 = \sqrt{\mu_0/\varepsilon_0}$ (Ω)
recommended α	$1.256\,637\,061 \times 10^{-6}$	$8.854\,187\,82 \times 10^{-12}$	376.730 313
the reading	$1.256\,863\,683 \times 10^{-6}$	$8.852\,591\,34 \times 10^{-12}$	376.798 253

And what is this: the product. $1/\sqrt{\mu_0\varepsilon_0}$ is 299 792 458 m/s from either row; the reading leaves the speed of light exactly where the convention put it, because it moves the permeability up and the permittivity down by one factor. *And what is that*: the ratio. The impedance of free space, $Z_0 = 2\alpha h/e^2$, is where the whole of the reading's difference lands: the vacuum resists a wave 1.80 parts in ten thousand more than measured. The speed is the declared constraint and the impedance its measured price, the multiplier and the penalty of Section 6. *Proved here* as arithmetic from the reading and the declared constants; that the speed is the multiplier and the impedance its price is *proposed*. Under the other common convention, the permittivity declared one, the same difference appears instead as an elementary charge 0.9 parts in ten thousand small. The convention chooses where the residue is written, never whether it is there, and the impedance, measured to about one part in 10^{10}, excludes it as the anomalous moment does.

12 What the knower carries

Collected, the proposals of this card are five sentences. The search by the first variation is the principle of least action. The search by the derivative is electrodynamics. The inferred operator is the map between the field spaces. What lies past the meter's bound is the fifth busy beaver. The search carries the quarter, and the quarter weighs one part in ten million. None is a step; each is about the continuation of a record of which this card has only a prefix.

The prefix is shown or cited in full: that the affine term is an increment and its derivative a multiplier; that the scalar derivative is exact with remainder $13h^2$ and squares to the discriminant at the reading; that the jar is the lower wall and its first Newton image; that both searches print one integer; that a reader through the card between reads the device's number when it deducts the slot, and carries the slot when it does not. Every Pythagorean text in the canon is used once, at a named line. A reader who refuses one knows which line falls, and a reader who accepts all of them is still handed the five sentences graded, so that they are carried as texts and not mistaken for receipts.

The argument closes on the bent card (*The Infinite Loop Considered Harmless*, §5). The record needs three receipts it cannot write for itself: the base case of its search, which Prop. 9.4 supplies; the zero its halting test declares; and the constants that give it units, under which the reading becomes the impedance of free space. Each is a fixed point handed in from outside, and none absorbs the residue.

The last sacred text. The canon of Section 2 lists the texts the derivation stands on, and Section 4 follows them down to the electron. One more is not on either list, because the device cannot cite it: it is the text the reader brings. The device shows that its own constructions do not close the gap. Every reader that measures length, every slot, every free seat, both cuts, and two

machines leave the reading where it is, and no declaration absorbs the residue. *Measured.* It does not show that no construction anywhere does. Passing from *not constructed here* to *does not exist* quantifies over every possible construction, which no finite search reaches; it is the step the halting test declares and the step from a double negation to an assertion, and it is taken by belief.

So the last sacred text is a sentence and the belief that travels with it: *the device's failure to construct the gap demonstrates that there is none.* It is Pythagorean in the sense of Def. 1.1, used at one place only, the residue, and never shown. The reader's question is the Quarter's: can you find the polynomial in the physical model of this device? The knower's choice is whether this text is true. Everything else on this card is a receipt.

■

A Sources

This appendix is the only part of the card that names the middle card's files and declarations. Paths are relative to `CARD_2/Measurement/`; declarations are cited by name. The working tree was inspected on 31 October 2026; nothing here reports a build.

This card	Middle card (declaration)
Def. 1.1	`Episode04.lean`: `Cult` (`inside_joke`, `pythagoras`, `triangles`), `InitiationProcess.sacred_texts`; `Episode09.lean`: `STEP25_BUILD_LOCAL_GEOMETRIES_THAT_SATISFY_THE_TRIANGLE_INEQUALITY`; `Episode10.lean`: `REVIEWER_2_ASKS_IF_DEVOTION_TO_METHOD_IS_A_METHOD`
Epigraph	`Episode04.lean`: margin of `the_experiment`
§3	`Episode10.lean`: `REVIEWED`; `Episode11.lean`: `TYPESET`, `pressCheck`
Prop. 8.2	`Episode12.lean`: `jarLo`, `jarHi`, `theLowWall`, `theHighWall`
Prop. 8.5	`Episode12.lean`: `halve`, `frechet_walk`, `theBisectionLedger`, `theNewtonLedger`, `the_bisection_ledger_turns`, `the_newton_ledger_turns`, `ledger_is_the_fold`, `the_crank_is_its_ledger`, `the_bracket_is_its_ledger`, `the_low_wall_is_above_the_vertex`, `the_printed_reading_is_the_natural`, `the_division_was_quot_sound`, `what_would_you_call_this_number?`
Def. 9.1	`Episode08.lean`: `bent_card.the_device_computing_the_next_configuration`, `infer`, `the_truth_inferred_by_the_device`
Carriers	`Episode01.lean`: `CarrierProcess` (`symbol`, `quarter`, `interact_with_the_universe`), `FACT.Truth`; `Episode08.lean`: `Number.peano`; `Episode12.lean`: `the_card_between`, `the_mark_between`, `the_tape_between_deals_rung01`, `deal_rung`, `upness_seats`
§10, pointwise exact	`Episode03.lean`: `Polynomial.le` and `Spline` with their Weierstrass margins; `Episode04.lean`: `Variation.le`, the `the_frechet` arm and its margin (same position, same velocity, same acceleration)
Def. 9.6; two readings	`Episode01.lean`: `Natural.le` and the rational order with its covariant/contravariant margin; `Episode02.lean`: `PigeonHole` order (superpigeons, pigeons, boxes); `Episode12.lean`: `same_click`, `the_chain_closes`, `THE_ACCOUNT_CLICKS`, `the_closure_between`, `SAME_DIFFERENCE`, `the_click_diagonal`
Prop. 9.4	`Episode12.lean`: `the_other_machine_agrees`, `relettering_is_free`, `the_clicks_read_the_number`

This card	Middle card (declaration)
The search for the quarter	`Calibration/LeanCalibration.lean`: `EKG`, `EKG.outgrown?`, `busyBeaverBudget`, `register_noise_floor`; `Calibration/EKGBounded.lean`: `tryBounded?`; `Episode05.lean`: `teraFLOPs`, `trilinos`, `HALTED`
§11	`Episode11.lean`: `TYPESET`, `PressVerdict`, `pressCheck`

B The sealed predictions

Written on 31 October 2026, before the run that decides them: the middle card built from its working tree on a second machine, then a three-reader comparison through the card between (the carrier's own interaction, the search reader, and the exact reader of Def. 9.3).

- The exact reader reproduces 137.011290548979457087737300 to every place, bills symbol identity only, and agrees by definitional unfolding with the middle card's pair.
- The search reader finds the reading displaced by about one part in 10^7.
- Both lie in the jar $[129.6, 137.7]$.
- The carrier's own interaction, applied to a tape folded onto a zero, keeps its first symbol and the slot, so every nonempty tape reads two cells and the reading it feeds the search is degenerate.

The run's outcome is to be entered here, beside each line, and not anywhere else in the card.

Outcome, 31 October 2026. The middle card built cleanly on the second machine (twenty-two minutes; peak resident memory 1.56 GB). The three-reader comparison was then started on one core and stopped by hand after seventy minutes, with its memory footprint at 92 GB on a machine of 64 GB and the excess in compressed memory and swap. It printed no reading. A first attempt on a machine of 8 GB had been stopped at 7 GB for the same reason. *All four predictions above remain undecided.* The first is not thereby weakened: it follows from Prop. 9.4 on paper, and the run was its machine check, not its proof. The second, the size of the displacement, has no other support and waits on a run that fits. Separately, on the same day and machine, every free universe seat of the ladder's rungs fourteen to thirty-eight was moved off its computed pin; all twenty-five rungs dealt, clicked exactly as the pinned rungs do, and billed the same receipts. *Measured.*

Second run, the same day. The comparison was repeated in four stages, each guarded at 16 GB and fifteen minutes, with the walk read from where the fact card stores it instead of being recomputed.

- *Exact reader: confirmed.* Its pair equals the middle card's pair digit for digit, and its reading is 13701129054897945708773730 0 at scale 10^{24}, the device's own, in under a second. The receipt is proved: folding a tape onto a slot appends the slot (no axioms); the exact reader is the length (symbol identity); the reading through the card between is the middle card's (symbol and value identity, the latter from function extensionality, so the receipt costs one value identity more than the first line above predicted; the reading itself costs none).

- *Carrier's own interaction: confirmed.* Its pair is $(0, 0)$: every tape reads two cells, the start lands on a double root, and the first step sends it to zero.

- *Search reader: no reading.* Stopped by the guard at 18 GB after eight minutes, in compiled evaluation. The measured lengths explain it. The fact card's step has $D = 13$, $P = 18$, $B = 2$ and starts at $(530, 324)$; this is N exactly (Prop. 9.2), and on ratios it halts at its seventh turn on the certified integer, which a ratio model of the walk reproduces to every place. The search reader adds the slot's one cell to every length: $D = 14$, $P = 19$, $B = 3$, and the step no longer divides by the derivative. Its fixed points are $(57 \pm \sqrt{1121})/56$; the upper one, the counterpart of the reading, repels, with multiplier about -2.08. The halting test never holds in 107 turns, so the unnormalized pair squares its size every turn, which no machine can store. On ratios the walk wanders and ends at a reported value near 220.10, outside the jar.

Measured for the lengths, the pairs and the stopped run; *computed*, from the measured lengths, for the ratio model. So the first prediction holds, the fourth holds, the third holds only in the sense that the jar is built from the middle card's own lengths (the search reader's walk leaves it), and the second, a displacement of one part in 10^7, is not what the search reader through the card between does: it finds no nearby number at all, because the carried slot turns the derivative's step into one whose fixed point repels. Whether some other search, one that carries the quarter without breaking the derivative, displaces the reading by that amount is open.

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